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INDEPENDENT RESEARCH / WORKING PAPER

Onchain Transparency Is Not Proof

*A Forensic Evidence Model for
State, Constraint, and Authority in DeFi*

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A bounded study in financial evidence reconstruction for consequential onchain claims.

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Source boundary. Publicly disclosable findings from Experiments 001A through 020 using bounded Aave V3 Ethereum observations at block 20,000,000. Credentials, reusable extraction procedures, execution logic, private commercial methods, and unsupported actor-specific authorization conclusions are excluded.

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Abstract

Onchain finance creates an illusion of evidentiary completeness. Events are public, balances are queryable, risk outputs are precise, and governance transactions are permanent. Yet these records do not prove the same facts. This study tested the problem through twenty bounded experiments on Aave V3 Ethereum at block 20,000,000. Event-derived account reconstructions contradicted exact-block state. Aggregate debt failed to identify reserve-level obligations. A measured health-factor distance required separate oracle, reserve-configuration, coverage, arithmetic, and provenance evidence before it could be explained. Historical configuration actions divided into proposal-governed and delegated execution paths whose authorization chains were structurally different. The result is an evidence model that separates economic relation from evidence metadata and distinguishes observed state, derived measurement, bounded explanation, historical action, and authority provenance. The study is bounded, not protocol-wide. Its contribution is a falsifiable warning: transparent data can still support materially overclaimed financial conclusions when evidence classes are collapsed.

PRINCIPAL FINDING. Public data is not the same as proven state. A numerically correct output can remain causally unexplained, and a successful transaction can remain authorization-incomplete. Transparency does not remove representation risk.

1. Executive Summary

DeFi is often described as radically transparent. The description is correct and dangerously incomplete.

The ledger exposes transactions, events, token balances, contract code, and protocol calls. This abundance creates a recurring analytical shortcut: if the data is public and the number is reproducible, the conclusion is treated as proven. Our experiments show why that shortcut fails.

At a fixed Ethereum block, event-derived reconstructions of Aave V3 account positions disagreed with direct historical contract state. Account-level aggregate debt established that debt existed but did not identify the reserve carrying it. A health factor could be observed exactly and its distance from liquidation measured, while its causal explanation remained incomplete until oracle prices, reserve configuration, decimals, liquidation thresholds, reserve coverage, and protocol arithmetic were reconstructed separately. Historical configuration actions could be located onchain, yet the authority behind those actions required a further chain through proposals, payloads, executors, role grants, stewards, automation registrations, and mandates.

The failure is not that onchain data is unreliable. The failure is treating different evidence objects as interchangeable because they refer to the same economic relationship.

The financial evidence reconstruction method developed in this study addresses this by requiring every statement to contain two separable parts:

economic_relation + evidence_metadata

The economic relation states what is alleged: an account holds a claim, owes a resource, faces a constraint, or possesses authority. Evidence metadata states how that allegation is supported, at what block, through which method, with what derivation status, coverage, confidence, contradiction, and Unknown.

The implication is direct. Any institution relying on onchain evidence for risk, governance, treasury, compliance, underwriting, or incident review should be able to answer five questions before calling a claim proven:

1. What exact economic relation is asserted?
2. Which evidence class supports it?
3. What state, time, and coverage boundary applies?
4. What was observed, derived, inferred, contradicted, or left Unknown?
5. Does the authorization chain support the conclusion, or only the fact that execution occurred?

If those questions cannot be answered, the analysis may still be useful. It is not yet a complete evidence record.

2. Problem Definition

2.1 Transparency Produces Confidence Faster Than Proof

Onchain systems make verification easier than conventional finance in important ways. Transactions are durable. Contract calls are reproducible. State can be queried at exact historical blocks when archive access exists. Token representations can be cross-checked. Governance execution can be traced.

But visibility does not determine meaning. A transfer event records a transition, not necessarily the complete position after all transitions. A debt aggregate reports a total, not necessarily its reserve composition. A health factor reports protocol-computed pressure, not the evidence chain explaining that pressure. A successful call proves execution, not necessarily the governance mandate or delegated role that permitted it.

This creates **representation risk**: the risk that an analysis is precise, reproducible, and materially wrong about what its evidence proves.

2.2 Four Claims Commonly Collapsed

The research repeatedly encountered four statements treated as though they were equivalent:

Claim	Evidence actually required
An event happened	historical transaction or event evidence
A state existed	exact-block state observation with defined coverage
A constraint was explained	state inputs, protocol rules, arithmetic, and provenance
An action was authorized	proposal, payload, executor, role, delegation, or mandate evidence

One layer may support another. None automatically proves the next.

2.3 Research Question

Can a bounded DeFi lending market be represented in a way that preserves state, history, derivation, explanation, authority, contradiction, and Unknown without collapsing them into one confidence-producing data view?

3. Methodology

3.1 Bounded Specimen

The empirical specimen was deliberately narrow:

- **Protocol:** Aave V3.
- **Chain:** Ethereum.
- **Observation block:** 20,000,000.
- **Initial specimen:** selected account-reserve pairs.
- **Repaired account coverage:** all 30 initialized reserves for one zero-debt account.
- **Scaled specimen:** 12 stratified accounts and 360 reserve observations.
- **Primary state method:** exact-block historical `eth_call`.
- **Action boundary:** research only; no trading or execution output.

The work does not claim exhaustive Aave coverage or universal DeFi validity.

3.2 Evidence Classes

The experiments separated at least the following evidence classes:

- **Frozen State Observation:** exact contract state at the target block.
- **Account-Level Aggregate State:** protocol-computed totals and risk outputs.
- **Reserve-Level Direct State:** account-reserve claim, obligation, and collateral state.
- **Token-Level Representation:** aToken and debt-token balances.
- **Event Observation:** historical supply, borrow, liquidation, configuration, or execution evidence.
- **Derived Measurement:** a value computed from observed inputs.
- **Bounded Explanation:** a reproduced protocol output supported by covered inputs and rules.
- **Authority Provenance:** evidence connecting action to proposal, payload, executor, role, registration, delegation, or mandate.
- **Inference:** interpretation not directly returned by the observed system.
- **Unknown:** evidence required by the claim but not established.

3.3 Economic Representation

The working economic ontology remained compact:

- Agent
- Resource
- Claim
- Obligation
- Constraint
- Authority

No experiment through Experiment 020 forced an additional economic primitive. More detailed structures, including tokenized claims, debt tokens, graph nodes, evidence classes, confidence vectors, and provenance chains, survived as relationships, implementation mechanics, derived views, or evidence metadata.

3.4 Research Sequence

Figure 1 shows the sequence. Each arrow represents a question that could not be answered safely by the preceding layer alone.

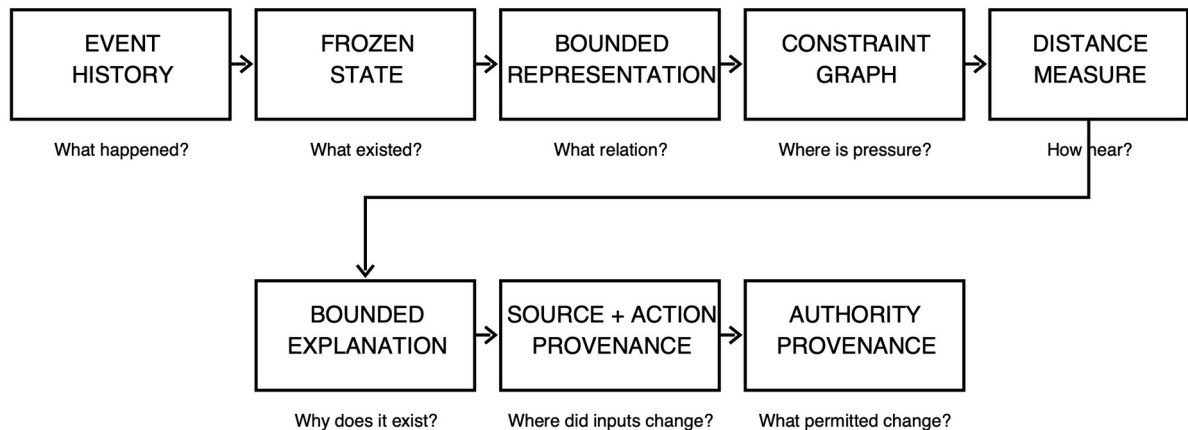


Figure 1. Evidence reconstruction sequence

The sequence is not a software architecture claim. It is the order in which evidence failures became observable.

4. Evidence

4.1 Event History Failed As Frozen State

The corrected first experiment used event-derived supply, borrow, liquidation, and net-position files to reconstruct Aave relationships up to block 20,000,000. Those files supported partial historical reconstruction. They could not prove complete user reserve state, collateral flags, token balances, oracle state, or authority at the exact block.

Archive-capable historical calls then returned direct account and reserve data. For selected pairs, direct reserve state matched aToken, stable-debt-token, and variable-debt-token balances while contradicting earlier event-derived reconstructions.

The result is stronger than “events can be noisy.” Event history and frozen state are answers to different questions. Agreement must be demonstrated; it cannot be assumed.

4.2 Aggregate Debt Hid Reserve Composition

Two selected accounts returned positive account-level debt while initially selected reserve-level reads returned zero debt. The aggregate established an obligation, but not its resource composition.

A reserve expansion found the actual variable-debt reserve: WETH. Direct `getUserReserveData` values matched variable-debt-token `balanceOf` outputs. One historical liquidation path pointed to MKR while current exact state showed WETH debt, creating a contradiction. Another historical WETH liquidation path matched current WETH debt, creating a confirmation.

The same event class therefore produced opposite relationships to current state. Historical relevance had to be represented, not presumed.

4.3 The Constraint Surface Was Mechanically Detectable

Three validated account representations were combined into a bounded surface and graph. The graph preserved agents, resources, claims, obligations, collateral constraints, health factors, evidence classes, Unknowns, contradictions, and confirmations.

Mechanical detection found:

- WETH as the positive obligation cluster;
- DAI as the largest claim cluster by account count;
- two finite near-threshold accounts;
- one zero-debt collateral account;
- one event-versus-state contradiction;
- one event-versus-state confirmation;
- Unknown oracle and authority state attached to accounts.

These were research classifications, not liquidation priorities. The distinction mattered because the graph could locate pressure before the research could explain it.

4.4 Distance Was Observable Before It Was Explainable

Using the Aave liquidation boundary of 1.0, the study calculated:

$$\text{distance_to_boundary} = \text{observed_health_factor} - 1.0$$

The two finite debt-bearing accounts had approximate buffers of 5.65 percent and 20.32 percent. The zero-debt account returned the protocol maximum and was classified as `unbounded_by_current_debt`, not ranked against finite accounts.

At this stage, observation confidence was High while explanation confidence was Low and action confidence was Not Assessed. The numbers were valid measurements. Their causal evidence chain was incomplete.

4.5 Bounded Explanation Required Exact Inputs

Experiment 009 retrieved exact-block Aave oracle prices, reserve decimals, LTV values, liquidation thresholds, reserve indexes, account aggregates, and represented reserve balances. Using protocol arithmetic, it reproduced the Pool health-factor outputs for the two finite accounts.

This changed the status from merely observed distance to bounded explained distance. It did not establish deeper provenance. Underlying oracle rounds, governance configuration lineage, and one account's full reserve coverage remained separate questions.

This distinction prevents a common error: treating a reproduced formula as a complete causal account. A formula explains the output only within the provenance and coverage of its inputs.

4.6 Oracle Prices Were Evidence Chains

For WETH, direct AggregatorV3-compatible round data supported the source output behind AaveOracle. For DAI and USDC, capped wrapper outputs and underlying aggregator rounds matched, but wrapper-level round mechanics remained incomplete.

The price evidence therefore had different provenance depth even where the observed AaveOracle output was exact. A uniform "oracle verified" label would have destroyed this distinction.

The provenance expansion later covered ten additional resources. Exact AaveOracle prices and reserve configurations were observed for all ten. Source provenance was High for three and Partial for seven. Seven of twelve scaled accounts gained improved price, configuration, or source-layer confidence, while historical authority provenance remained unresolved.

4.7 Full Coverage Changed Confidence, Not Ontology

One zero-debt account had initially been represented through a selected DAI pair. A full scan of all 30 initialized reserves confirmed DAI as its only positive claim, no stable or variable debt, collateral-enabled DAI, maximum health-factor behavior, and non-comparable distance status.

The account representation did not require a new field or primitive. What changed was the evidence supporting completeness. Selected-reserve absence and full-universe absence are not the same claim.

4.8 Scaling Preserved Evidence Classes

A four-account dry run completed 120 reserve observations with zero reserve failures and zero token cross-check mismatches. The full scaled experiment then represented 12 stratified accounts across 360 reserve observations, again with zero reserve failures and zero token cross-check mismatches. Thirteen resources appeared in the sample.

Confidence vectors and Unknown vectors survived scaling. Action confidence remained Not Assessed for every account. The principal scaling cost was not schema failure. It was a wider provenance burden as new resources appeared.

4.9 Historical Action Was Not Authorization

For ten expanded resources, historical source-selection and reserve-initialization actions were recovered for all ten. Collateral and reserve-factor actions were recovered for nine; borrowing configuration for all ten; stable-borrowing action for four. Governance authorization was initially Unknown for all ten resources.

Experiment 019 reviewed 17 unique latest-action transactions. Seven linked through executed Aave Governance V2 proposals, payload targets, the short executor, execution transactions, and resulting configuration actions. Ten followed delegated steward or automation paths.

The split is important. A proposal-only authority model would have misrepresented valid delegated mechanisms. An execution-only model would have overstated authorization.

4.10 Delegated Execution Had Multiple Evidence Links

The final authority experiment examined the ten delegated paths. Two were verified one-shot WBTC and wstETH listing stewards with listing-engine dependencies and pool-admin checks. Proposal 147 explicitly granted POOL_ADMIN to each listing steward and required renunciation after the one-asset listing, achieving the bounded authorization chain for those two paths.¹ Seven actions passed through the Chainlink Keeper Registry. Exact-block getUpkeep calls recovered three downstream targets, action-specific

¹ Aave, "Aave Ethereum V3," Governance V2 Proposal 147, executed January 27, 2023, accessed July 11, 2026, <https://governance-v2.aave.com/governance/proposal/147/>.

administrators, and common Aave robot-operator infrastructure.² One weETH leverage() path remained authority-Unknown.

Immediate execution mechanics were explained for nine of ten paths. Originating authorization was achieved for two, partial for seven, and Unknown for one. This produces the paper's most consequential authority distinction:

execution != registration != administration != role grant != governance mandate

A permanent transaction record proves that execution occurred. It does not automatically prove every preceding authorization link.

5. The Evidence Reconstruction Model

5.1 Statement Structure

Every consequential statement should identify:

- **Economic relation:** subject, predicate, object, quantity, resource, and ontology mapping.
- **Evidence metadata:** evidence class, observation block, source method, derivation status, and confidence.
- **Coverage:** account, reserve, protocol, chain, time, and universe boundary.
- **Provenance:** upstream source, configuration, authority, and transformation links.
- **Exceptions:** Unknowns, contradictions, non-comparable states, and failed reads.

5.2 Proof Ladder

The experiments support a practical proof ladder:

1. **Observed:** a value or event was directly retrieved.
2. **Cross-checked:** an independent evidence representation agreed.
3. **Covered:** the relevant bounded universe was scanned or explicitly limited.
4. **Explained:** protocol output was reproduced from observed inputs and rules.
5. **Provenanced:** source and transformation lineage were established.
6. **Authorized:** proposal, payload, role, delegation, registration, or mandate supported the action.

Not every claim needs every rung. The required rung depends on the conclusion. A balance claim may require exact state and coverage. A causal risk claim requires explanation. A governance legitimacy claim requires authorization provenance.

² Chainlink, "Chainlink Automation," documentation, accessed July 11, 2026, <https://docs.chain.link/chainlink-automation>.

5.3 Unknown Is A Result

Unknown does not mean no evidence exists. It means the evidence required for the stated conclusion was not established within the bounded method.

Preserving Unknown prevents three distortions:

- absence of proof becoming proof of absence;
- successful execution becoming inferred authorization;
- observed output becoming inferred causal explanation.

6. Representation Risk

Representation risk is operational, not philosophical.

For a risk committee, collapsing account aggregates and reserve composition can hide the resource carrying an obligation. For governance review, collapsing action and authorization can produce an unsupported legitimacy claim. For treasury or underwriting, collapsing observed price and source provenance can disguise dependence on wrappers or configuration choices. For automated systems, collapsing observation confidence and action confidence can turn a valid measurement into an unjustified instruction.

The dangerous output is not always a wrong number. It is a correct number attached to a stronger claim than its evidence supports.

EVIDENCE BOUNDARY. This paper demonstrates representation failures and repairs inside a bounded Aave V3 Ethereum specimen. It does not determine that any named actor acted improperly, that Aave governance is invalid, or that the same empirical frequencies hold across DeFi.

7. Implications

7.1 Protocol Risk And Governance

Risk reports should distinguish current configuration, the historical action that produced it, and the authority path permitting that action. A single governance label is insufficient when protocols use proposals, executors, guardians, stewards, keepers, and bounded automation.

7.2 Audits And Incident Reviews

Transaction traces should not substitute for exact-state reconstruction where the claim concerns state. Conversely, current state should not erase historically material actions. Both should coexist in the evidence graph.

7.3 Financial Intelligence Systems

A credible intelligence engine should return not only a value but also its evidence class, coverage, derivation, contradictions, provenance depth, and Unknowns. Confidence must describe evidence quality, not silently become action priority.

7.4 Commercial Decision Review

The same model applies beyond protocol mechanics. A consequential financial decision combines claims, evidence, transformations, authorities, constraints, and unresolved Unknowns. The commercial value is not another dashboard. It is a decision record that shows exactly what is proven, what is merely observed, and what remains unsupported before capital or authority is committed.

8. Limitations

The empirical work used one protocol, one chain, one historical block, and bounded account populations. The 12-account sample tested scaling discipline, not statistical representativeness. Resource concentration findings are specimen findings, not market estimates.

Exact-block calls depend on archive-capable infrastructure. Although independent token representations and repeated calls supplied cross-checks, provider-level trust and client implementation were not eliminated.

Bounded health-factor explanation was achieved for two finite accounts. It does not prove that every Aave account or every protocol risk metric can be reconstructed with the same coverage.

Oracle-source provenance remained partial for wrappers and custom sources. Full offchain data-generation provenance was outside scope.

Delegated authority provenance was achieved for two reviewed paths, remained Partial for seven, and Unknown for one. The paper therefore makes no unsupported actor-specific claim of unauthorized conduct, improper delegation, or governance failure.

No profitability, liquidation eligibility, transaction execution, or action confidence was assessed. The work does not establish that any detected constraint should trigger intervention.

8.1 Unknown Register

Material public Unknowns include:

- full originating role-grant chronology for seven selected automation paths;
- formal mandates and registration transactions for bounded automation paths;
- the authorization chain behind the `weETH leverage()` path;
- full wrapper-level oracle mechanics for selected capped or custom sources;
- cross-protocol and cross-chain generality;
- production-scale performance beyond the controlled 12-account specimen;

- measured commercial decision impact.

9. What This Paper Does Not Claim

This paper does not claim:

- that events are useless;
- that exact state alone explains causation or authority;
- that Aave produced false data;
- that any account should be liquidated;
- that any governance participant acted without authority;
- that the bounded sample represents all Aave activity;
- that this study has proven a universal financial ontology;
- that confidence classifications are action signals;
- that commercial value has been proven by a paid engagement;
- that protected extraction or execution methods are disclosed.

10. Conclusion

Onchain finance is transparent enough to reveal its own evidence failures.

The bounded Aave V3 program began with an apparently straightforward reconstruction and found that event history could not prove frozen state. Direct state then revealed that account aggregates could not identify reserve obligations. Graphs located pressure before causal inputs explained it. Oracle and configuration reconstruction produced bounded explanations while exposing deeper provenance gaps. Historical actions split into proposal-governed and delegated paths. Delegated execution exposed additional links between registry, target, administrator, role, and mandate.

The resulting principle is severe but useful:

A financial claim is only as strong as the weakest evidence transition it silently asks the reader to accept.

Public ledgers do not remove that transition. Dashboards often hide it. Precision can intensify it. The method makes it explicit.

The next generation of onchain intelligence should not merely collect more data. It should reconstruct what each item proves, preserve what it does not prove, and refuse to turn visibility into certainty without an evidence chain.

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Appendix A. Reproducibility Statement

The canonical public evidence manifest identifies every source artifact supporting a public claim. Exact account addresses and bounded values already present in public-source materials may be independently queried at Ethereum block 20,000,000 using archive-capable RPC infrastructure and standard Aave V3 interfaces. Credentials are not included.

The public package includes the experiment index, claim-to-evidence matrix, limitations and Unknown register, artifact hashes, and release-review records. Reusable private extraction procedures and operational credentials remain excluded. This limits one-click replication of the private pipeline but does not change the stated evidence boundary.

Appendix B. Controlled Access Statement

This paper is independent research by Darwin Rampai. It discloses a bounded methodological record in financial evidence reconstruction. Full internal artifacts, reusable implementation procedures, and extended applications remain controlled. Private briefings and commissioned reviews may be considered under terms that do not alter evidence classifications, Unknowns, or determinations.